

Weekly Progress Report: SLM-to-SLM Guided Reasoning — Controls, Ceilings, and Cross-Architecture Validation

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Abstract—This weekly report documents the new experimental work completed since Week 4. Four major contributions are reported: (1) completion of the zero-shot chain-of-thought (CoT) baseline at equal compute across five datasets, yielding a three-tier task taxonomy; (2) completion of the 3B ceiling baseline, formalising a two-pronged elimination argument that isolates fine-tuning as the sole causal mechanism; (3) task-specific fine-tuning experiments on Qwen2.5-3B and Microsoft Phi-3.5-mini-instruct as cross-architecture validation; and (4) a fine-tuning data quality ablation comparing ASDiv-trained versus GSM8K-trained guides. The core finding is strengthened: fine-tuning quality is the necessary and sufficient condition for effective SLM-to-SLM guidance, and the operative dimension of quality is *verbal plan richness and structural diversity*, not task-type domain overlap.

Index Terms—small language models, LoRA fine-tuning, guided reasoning, chain-of-thought, ceiling baseline, cross-architecture, fine-tuning data quality, task taxonomy

I. PROJECT OVERVIEW

The ongoing research investigates whether a fine-tuned SLM can replace expensive LLM guides in a guided reasoning pipeline. A LoRA fine-tuned [6] Qwen2.5-3B guide generates structured verbal plans—drawing on chain-of-thought principles [1]—consumed by a Qwen2.5-1.5B solver ensemble using 5-vote majority voting [2]. Week 4 established statistically confirmed accuracy improvements on four datasets. This week addresses the three remaining control conditions: CoT prompting, raw model scaling, and alternative fine-tuning configurations.

II. COMPLETED WORK THIS WEEK

A. ARC-Challenge Scaled to $N=900$ and Confirmed

ARC-Challenge was underpowered at $N=100$ in Week 4. Extended to $N=900$: McNemar’s test [10] yields $\chi^2=26.71$, $p<0.0001$ (★★★★). The accuracy gain stabilises at +8.3 pp (baseline 71.7%, guided 80.0%), confirming the $N=100$ result was genuine but underpowered.

B. CoT Baseline Control Completed

The zero-shot CoT baseline [4] at equal compute (7.5B param-passes) was run across five datasets (Table I), yielding a three-tier task taxonomy.

Three-Tier Task Taxonomy.

TABLE I
THREE-WAY ACCURACY COMPARISON: BASELINE VS. CoT VS. GUIDED.

Dataset	N	Base	CoT	Guided	Guide—CoT
SVAMP	300	36.7%	47.0%	61.7%	+14.7
PIQA	500	55.4%	54.8%	78.4%	+23.6
CSQA	900	70.4%	67.7%	75.8%	+8.1
ARC-C	900	71.7%	80.0%	80.0%	+0.0
ASDiv	300	47.6%	45.8%	64.0%	+18.2

- **Tier 1 — Retrieval (RACE-High):** Neither CoT nor the guide helps; verbal planning offers no leverage.
- **Tier 2 — Domain-Accessible (ARC-Challenge):** CoT is sufficient; both reach 80.0%. The guide adds zero marginal value.
- **Tier 3 — Domain-Inaccessible (PIQA, CSQA, SVAMP, ASDiv):** CoT is insufficient or harmful. Fine-tuned verbal planning delivers +8–24 pp that CoT cannot replicate.

On PIQA, CoT corrects position bias but accuracy remains 54.8% vs guided 78.4%—a +23.6 pp gap—showing debiasing alone is insufficient. On CommonsenseQA, CoT drops accuracy −2.8 pp and worsens ECE (0.191), while the guide achieves +8.1 pp and ECE 0.141.

C. 3B Ceiling Baseline Completed

The untuned Qwen2.5-3B with 5 votes (15.0B param-passes) was evaluated as a size-matched ceiling (Table II). On arithmetic tasks, the untuned model’s verbose outputs fail format extraction, collapsing below baseline. On ARC-Challenge, the 3B ceiling (66.8%) falls below the 1.5B baseline (71.7%). The guided pipeline equals or exceeds the ceiling on all five datasets at 30% lower compute (10.5B vs 15.0B param-passes).

Two-Pronged Elimination Argument.

- **Prong 1 (Ceiling):** Size alone is insufficient and can be harmful.
- **Prong 2 (CoT):** Prompting alone adds zero or negative accuracy on four of five Tier 3 datasets.

Both alternatives eliminated $\Rightarrow$ **fine-tuning quality is the necessary and sufficient mechanism.**

TABLE II
CEILING BASELINE RESULTS.

Dataset	Ceiling	Guided	Guide—Ceil	Note
ARC-C	66.8%	80.0%	+13.2	Ceiling below baseline
PIQA	75.2%	78.4%	+3.2	Scale helps; FT wins
CSQA	75.0%	75.8%	+0.8	Tied; guided 43% cheaper
SVAMP	30.7%	61.7%	+31.0	Format extraction collapse
ASDiv	28.7%	64.0%	+35.3	Same format collapse

D. FT Qwen2.5-3B Solo Baseline

A LoRA fine-tuned [6] Qwen2.5-3B (rank-8, 29.9M trainable params, 0.96% of total) was evaluated at a single greedy pass (3.0B param-passes), $3.5\times$ lower compute than the guided pipeline (Table III).

TABLE III
FT QWEN2.5-3B SOLO VS. GUIDED PIPELINE.

Dataset	Base	FT Solo (3B pp)	Guided (10.5B pp)	Δ
ARC-C	71.7%	79.2%	80.0%	−0.8
ASDiv	53.7%	68.0%	64.0%	+4.0
CSQA	70.7%	80.6%	75.8%	+4.8
PIQA	54.2%	83.4%	78.4%	+5.0
SVAMP	40.3%	57.7%	61.7%	−4.0
Average	58.1%	73.8%	72.0%	+1.8

The FT Qwen 3B solo averages 73.8% vs the guided pipeline’s 72.0%, matching or exceeding it on three of five benchmarks at $3.5\times$ lower compute, raising an important cost-effectiveness question for the pipeline.

E. FT Phi-3.5-mini-instruct: Cross-Architecture Validation

The same LoRA (rank-8) protocol was applied to Microsoft Phi-3.5-mini-instruct (3.82B parameters, 3.1M trainable, 0.08% of total) to test architecture-agnosticism (Table IV).

TABLE IV
CROSS-ARCHITECTURE RESULTS: FT PHI-3.5 VS. FT QWEN VS. GUIDED PIPELINE.

Dataset	Base	FT Qwen	FT Phi-3.5	Guided	Phi—Guide
ARC-C	71.7%	79.2%	84.2%	80.0%	+4.2
ASDiv	53.7%	68.0%	65.7%	64.0%	+1.7
CSQA	70.7%	80.6%	81.2%	75.8%	+5.4
PIQA	54.2%	83.4%	85.6%	78.4%	+7.2
SVAMP	40.3%	57.7%	60.0%	61.7%	−1.7
Avg	58.1%	73.8%	75.3%	72.0%	+3.3

Phi-3.5-mini achieves 75.3% average, exceeding both FT Qwen (73.8%) and the guided pipeline (72.0%) at $3.5\times$ lower compute. The sole exception is SVAMP (−1.7 pp). Fine-tuning gains are confirmed *architecture-agnostic*.

F. Fine-Tuning Data Quality Ablation

A guide trained on ASDiv (sparse, one-sentence problems) was compared against the GSM8K-trained guide [7] (Table V).

The ASDiv-trained guide paradoxically underperforms on its own test set (57.3% vs 64.0%), showing domain proximity is a *weak predictor* of quality. The operative dimension is **verbal reasoning richness**: GSM8K’s multi-sentence narratives teach transferable structured scaffolds; ASDiv’s terse templates do not. Average gain collapses 46% (from +13.8 to +7.4 pp), operationalising quality as **structural diversity and verbal depth**.

TABLE V
GUIDE GAIN COMPARISON: GSM8K FT VS. ASDIV FT (PP OVER BASELINE).

Dataset	GSM8K FT Gain	ASDiv FT Gain	Reduction
PIQA	+24.2	+16.6	−31%
SVAMP	+21.3	+8.0	−62%
ASDiv	+10.3	+6.3	−39%
CSQA	+5.1	+3.6	−29%
ARC-C	+8.3	+2.7	−67%
Average	+13.8	+7.4	−46%

III. REFINED CENTRAL CLAIM

The guide is necessary when the task requires reasoning beyond what prompting can access. When domain knowledge is directly retrievable (ARC: CoT = Guide = 80%), CoT is sufficient. When it is not (PIQA, CSQA, SVAMP, ASDiv), fine-tuned verbal planning provides +8–25 pp that CoT cannot. Quality is operationalised as the structural diversity and verbal depth of the fine-tuning training distribution.

IV. REMAINING TASKS

- **Generalisation on Multiple Domains:** Evaluate the guided pipeline across a broader set of task domains to assess generalisation beyond current benchmarks.
- **Generalisation on Multiple Models:** Extend experiments to additional SLM architectures to further validate architecture-agnostic fine-tuning gains.
- **Error Analysis:** Systematic categorisation of guided-wrong/baseline-correct question types across Tier 3 datasets.

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